

# SRIKRUSHNA PINAKI BUDI

[srikrushna220@gmail.com](mailto:srikrushna220@gmail.com) | (413) 472-1308 | [GitHub](#) | [LinkedIn](#)

## Summary

Full-Stack Software Engineer with a passion for building scalable, efficient systems using Java and Spring Boot. I specialize in transforming complex business needs into clean, cloud-native solutions whether that's cutting data latency by 90%, securing sensitive user data, or automating manual processes. I thrive in collaborative environments and enjoy mentoring others while continuously learning and applying new technologies.

## Education

**University of Massachusetts Amherst | Master of Science in Computer Science | CGPA: 3.95** **2023 - 2025**

- **Courses:** Systems for Data Science, Comp & Network Security, Machine Learning, Neural Networks: Modern Intro, Database Design and Architecture, Advanced Algorithms, System Defense and Test, Distributed and Operating Systems

**Osmania University, Hyderabad, India | Bachelor's in Electronics and Communication Engineering | CGPA: 7.97 / 10** **2017 - 2021**

- **Courses:** Computer Prog. & Prob. Solving, Comp.Org. & Architecture, Data Comm.& Comp. Networking, OOP in Java, Data Struct. & Algo.

## Skills

- **Languages:** Java, Python, JavaScript/TypeScript, SQL, Bash
- **Frontend:** React, Redux, Angular, HTML5/CSS3, jQuery
- **Backend:** Spring Boot, Spring Security, Microservices, REST APIs, Node.js, Express.js, JSON-RPC
- **Cloud/DevOps:** AWS (Lambda, S3, EC2, Glue), Docker, Kubernetes, Jenkins, CI/CD
- **Data:** PostgreSQL, DynamoDB, MongoDB, Apache Kafka, Apache Flink, Spark, Snowflake, Power BI
- **Testing & Security:** JUnit, Mockito, Selenium, OAuth 2.0, GDPR, XSS/CSP, Pen Testing (Kali/Nmap)

## Work Experience

**University of Massachusetts Amherst | Graduate Teaching Assistant** **Sep'24 - Dec'24**

- Instructed and mentored 50+ graduate students in secure Java application development, covering encryption (AES, RSA), secure coding practices, and vulnerability mitigation aligned with OWASP guidelines.
- Designed and evaluated hands-on labs using Java-based penetration testing tools (Burp Suite, OWASP ZAP) and network analysis (Wireshark, Nmap) to simulate real-world threats like SQL injection and XSS.
- Developed and demonstrated Java security libraries for implementing HTTPS, certificate pinning, and secure session management in Spring Boot applications.
- Supported advanced coursework in distributed systems and network defense using Java socket programming, multithreading, and cloud-based virtual environments (AWS EC2, Docker).

**Staples US Retail, Framingham, Massachusetts | Software Engineer Intern** **May'24 - Aug'24**

- Led the development of a real-time inventory API using Java and Spring Boot, syncing data across 500+ stores and cutting stock discrepancies by 90%.
- Optimized checkout service performance by implementing Redis caching and refining Spring Boot APIs, reducing latency by 35% during high-traffic events.
- Built a data reconciliation service that audited Kafka event streams against transactional databases, resolving over 200 data mismatches per month.
- Modernized legacy data pipelines by developing scalable PySpark jobs on AWS EMR, improving ETL processing speed by 40% for millions of daily records.

**Phenom People Pvt. Ltd. | Product Development Engineer** **May'23 - Oct'23**

- Modernized the analytics pipeline from batch to real-time by developing a Java-based Apache Flink application on Kubernetes, reducing data latency by 90% and cutting infrastructure costs by 30%.
- Improved query performance for analytics dashboards by optimizing the Apache Pinot data model with Java-based time partitioning and columnar indexing strategies.
- Implemented comprehensive monitoring with Prometheus and Grafana to track pipeline health, API performance, and data quality in real-time.
- Migrated 150M+ records cross-region in Snowflake with zero downtime using Dockerized Python ETL. Ensured 100% data integrity via Great Expectations validation and 200+ pytest cases.

**Amazon Development Centre | Software Development Engineer** **Oct'21 - Apr'23**

- Built an Expense Reimbursement Portal (React/Spring Boot) with DynamoDB (GSIs) for user profiles and PostgreSQL for ACID-compliant transactions. Automated 80% of manual processes using AWS Step Functions (GDPR compliance) and achieved 85% test coverage (JUnit/Mockito).
- Enhanced Vendor Management Dashboard by integrating 6 Java Spring microservices to aggregate 12 months of transaction data, vendor hierarchies, and sub-vendor activities, improving data accuracy and API response times by 35%.
- Reduced XSS attack vectors by 90% through CSP implementation with nonce validation in React and Java Spring apps handling sensitive PII, while refactoring view controllers for secure script loading.
- Designed a real-time fraud detection system using DynamoDB Streams and AWS Lambda, deploying Scikit-learn models on SageMaker to analyze transaction patterns, reducing false positives by 25% and flagging suspicious activity with Prometheus alerts.

**Cognizant | Full Stack Development Intern** **Apr'21 - Oct'21**

- Engineered a full-stack task management dashboard using Java with Spring Boot for robust backend APIs and Angular for the dynamic frontend, improving team task tracking efficiency.
- Designed and secured RESTful APIs for all CRUD operations, implementing JWT-based authentication to manage user access control.
- Automated testing using JUnit and Mockito, reducing regression testing time by 50% and increasing code reliability.
- Gained practical experience in Agile development and CI/CD pipelines, contributing to the deployment of a scalable web application.

Projects

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Applied Machine Learning for Child Rescue   <a href="#">Link</a></b><br><b>Independent Study in collaboration with the Attorney General's Office Forensic Department and UMass Amherst</b> <ul style="list-style-type: none"><li>Developed a machine learning pipeline to detect unique faces and classify age groups in multimedia files for forensic child rescue operations.</li><li>Designed and optimized face detection and age classification systems using YOLOv8, Haar cascades, and custom-trained PyTorch models, achieving high accuracy in diverse datasets.</li><li>Automated E01 disk image handling and recursive file extraction with Arsenal Image Mounter for seamless multimedia analysis.</li><li>Enhanced processing efficiency through multi-threaded video parsing, reducing analysis time by 60%.</li><li>Created an intuitive GUI using Goody for law enforcement to interact with forensic tools easily.</li></ul> <b>Technical Stack:</b> Python, PyTorch, YOLOv8, TIMM, HuggingFace Transformers, OpenCV, Arsenal Image Mounter. | <b>Dec'24</b> |
| <b>Doctor Appointment Booking and Patient Tracking System   <a href="#">Link</a></b> <ul style="list-style-type: none"><li>Developed a full-stack healthcare management system using the MERN stack (MongoDB, Express.js, React.js, Node.js). The system facilitates doctor appointment scheduling and patient tracking, integrating a user-friendly frontend with a robust backend and MongoDB for efficient data management. Enhanced patient experience by streamlining scheduling processes and providing a unified platform for healthcare administrators.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Dec'24</b> |
| <b>Adaptive Portfolio Management and Stock Price Forecasting Using LSTM Neural Networks   <a href="#">Link</a></b> <ul style="list-style-type: none"><li>Developed LSTM model for stock price prediction, optimizing architecture through hyper-parameter tuning.</li><li>Designed and implemented a good trading algorithm, achieving high returns and a favorable Sharpe ratio during simulation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Dec'23</b> |
| <b>URL Parser   <a href="#">Link</a></b> <ul style="list-style-type: none"><li>Designed and Developed a Service which Parses URLs stores them and based on the Content Type of the URL.</li><li>It publishes to Different Kafka Topics based on the Content Type of the URL and these Topics are subscribed by Different Workers which validate the Content and can process the Data and parse internal URLs.</li><li><b>Tech Stack</b> - Spring Boot, MySQL, MongoDB, Apache Kafka ,Docker, Kubernetes, Git</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Dec'22</b> |

Achievements and Certifications

- IEEE Xtreme 14.0 - Global Rank 800 - [Link](#)
- Credly Badges - [Link](#)
- AWS Academy Cloud Foundations - [Link](#)
- Microsoft Certified: Microsoft Azure Fundamentals - [Link](#)
- Microsoft Certified: Introduction to Programming using Python - [Link](#)
- Data Science Professional Certificate - [Link](#)
- Applied Data Science Specialization - [Link](#)
- CodeChef Certificate in Data Structures and Algorithms - [Link](#)

Positions of Responsibility

|                                                                                                                                                                                                                                                               |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>The Voices of Data Sciences Conference at UMass Amherst   Head of Logistics Team</b> <ul style="list-style-type: none"><li>Orchestrated end-to-end operations for 100+ attendees, boosting engagement 30% through cross-functional coordination.</li></ul> | <b>Dec'24 - Mar'25</b> |
| <b>IEEE MVSr Student Branch Program   Committee Head</b> <ul style="list-style-type: none"><li>Organized and was part of various Technical Events, Workshops and Seminars. Coordinated TechnoVanza Annual Event.</li></ul>                                    | <b>Jun'18 - Jul'21</b> |